

Decent Coaching Classes

Std. VIII (SB) — Mind Your Math - I

Answer Key — Chapter 5 (Expansion Formulae) & Chapter 6 (Factorisation, up to 6.3)

Q.1

- (1) (b) $a + b$
- (2) $3ab^2$
- (3) (a) $(x + 3)(x + 4)$
- (4) b^2
- (5) (a) $a^3 - 3a^2b + 3ab^2 - b^3$
- (6) quadratic
- (7) (a) $a^2 + ab + b^2$

Q.2

- (1) $(x + 9)(x - 4) = x^2 + 5x - 36$
- (2) $(2y - 7)(2y + 3) = 4y^2 - 8y - 21$
- (3) $y^2 + 11y + 18 = (y + 9)(y + 2)$
- (4) $m^2 - 5m - 36 = (m - 9)(m + 4)$
- (5) $3x^2 + 14x + 15 = (x + 3)(3x + 5)$

Q.3

- (1) $(2p + 5)^3 = 8p^3 + 60p^2 + 150p + 125$
- (2) $(3x - 4y)^3 = 27x^3 - 108x^2y + 144xy^2 - 64y^3$
- (3) $(a + 2b + 3c)^2 = a^2 + 4b^2 + 9c^2 + 4ab + 12bc + 6ac$
- (4) $8x^3 + 27y^3 = (2x + 3y)(4x^2 - 6xy + 9y^2)$
- (5) $125p^3 - 64q^3 = (5p - 4q)(25p^2 + 20pq + 16q^2)$

Q.4

- (1) $(2p + 3q)^3 - (2p - 3q)^3 = 72p^2q + 54q^3$
- (2) $(99)^3 = (100 - 1)^3 = 1000000 - 30000 + 300 - 1 = 9,70,299$
- (3) $20x^2 - 26x + 8 = 2(2x - 1)(5x - 4)$